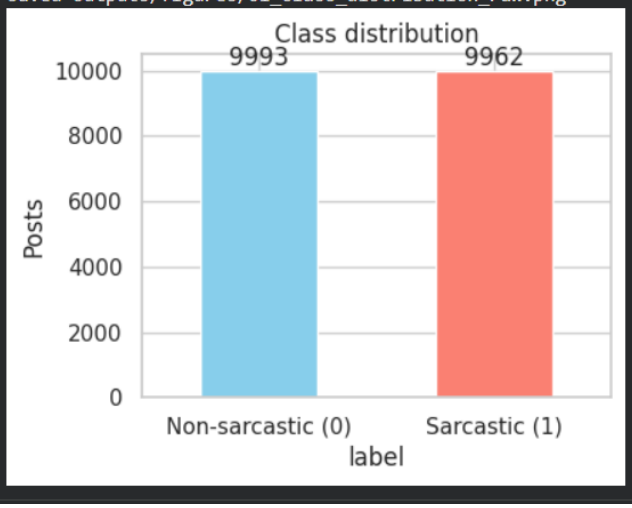
ID: 24K8016

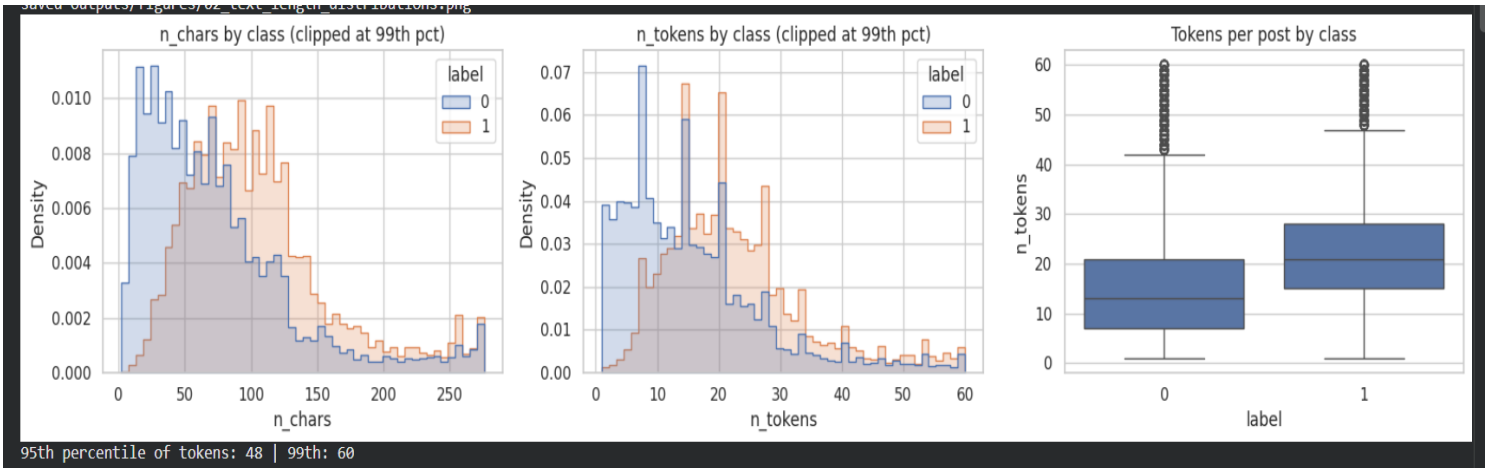
1.Exploatry analysis- Data preparation

1.1Report class distribution

```
saved outputs/results/eda_class_distribution_raw.csv  (2 rows)
saved outputs/figures/01_class_distribution_raw.png
```



1.2 Text-length statistics



label		0	1
n_chars	count	9993.00	9962.00
	mean	73.90	107.46
	std	58.97	57.93
	min	2.00	5.00
	25%	31.00	67.00
	50%	60.00	97.00
	75%	97.00	129.00
	95%	212.00	248.00
	max	327.00	304.00
n_tokens	count	9993.00	9962.00
	mean	16.09	23.09
	std	12.42	12.18
	min	1.00	1.00
	25%	7.00	15.00
	50%	13.00	21.00
	75%	21.00	28.00
	95%	43.00	51.00
	max	70.00	71.00
n_emoji	count	9993.00	9962.00
	mean	2.20	1.91
	std	2.63	2.13
	min	0.00	0.00
	25%	1.00	1.00
	50%	2.00	1.00
	75%	3.00	3.00
	95%	6.00	6.00
	max	71.00	42.00



1.3 Common data-quality issues

	overall_%	non-sarc_%	sarc_%
has_emoji	82.96	88.05	77.85
has_quote	4.18	3.20	5.16
has_excl_or_q	5.30	3.71	6.89
repeat_char	1.47	2.16	0.78
repeat_punct	13.93	13.70	14.16
has_hashtag	3.79	0.27	7.32
has_mention	3.87	0.83	6.92
has_url	1.90	0.32	3.48
has_latin	14.00	7.70	20.32
has_newline	4.81	0.78	8.85
foreign_chars>10%	0.54	0.99	0.09
very_short(<3 tok)	2.37	4.57	0.15

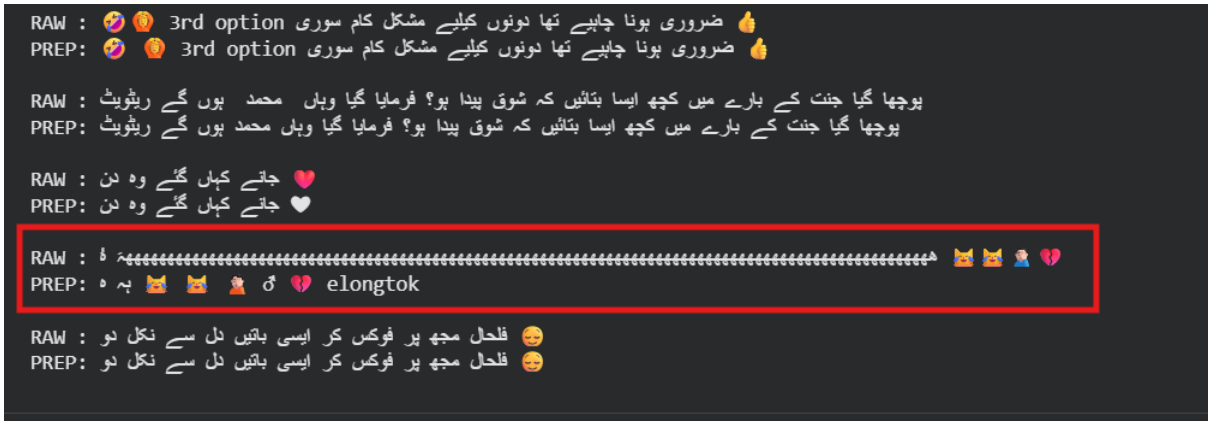
saved outputs/results/eda_quality_audit.csv (12 rows)

Exact duplicate rows: 7801 (39.09%)
Duplicate texts with CONFLICTING labels: 645
Empty / whitespace-only texts: 0

- **Duplicates:** 7,801 rows (39.09%) are exact duplicates. These were removed before splitting so the same post cannot appear in both train and test.
- **Conflicting labels:** 645 duplicate texts carry both labels. They are annotation noise, so they were dropped.
- Hashtags, @mentions, URLs and line breaks: all are much more frequent in sarcastic posts (for example, hashtags in 7.32% of sarcastic vs 0.27% of non-sarcastic posts), **so they were kept as cues.**
- Code-mixing: Latin-script words appear in 20.32% of sarcastic posts vs 7.70% of non-sarcastic posts, so English and Roman tokens were kept, **not removed.**
- Emojis: present in 83% of posts overall, and more common in non-sarcastic posts (88.05% vs 77.85%). Their usefulness depends on which emoji appears, so they are tested in the ablation.
- Repeated punctuation: about the same in both classes (14.16% vs 13.70%), so it is a weak cue.
- Very short posts (under 3 tokens): almost never sarcastic (0.15% vs 4.57%).

2 Urdu-aware preprocessing and cue analysis

2.1 normalization/tokenization



2.3 Do not blindly remove emojis, punctuation, repetition, hashtags, or code-mixed terms; test whether these are useful sarcasm cues.

cue	before non-sarc %	before sarc %	after non-sarc %	after sarc %
has_emoji	88.05	77.85	88.09	67.62
laugh_emoji	32.90	49.81	32.18	42.84
has_quote	3.20	5.16	3.10	5.20
has_exclaim	0.05	1.01	0.06	1.53
has_question	3.67	5.95	3.58	6.25
has_hashtag	0.27	7.32	0.28	11.48
has_mention	0.83	6.92	0.84	11.08
has_url	0.32	3.48	0.31	5.51
has_newline	0.78	8.85	0.82	14.07
has_latin	7.70	20.32	7.53	27.07
has_elongation	1.39	0.70	1.46	0.77
has_laughter	0.65	0.82	0.68	0.92
has_repeat_punct	14.27	14.46	14.39	14.67
very_short	4.57	0.15	4.82	0.23

	AUC_alone	distance_from_0.5
n_chars	0.722	0.222
n_tokens	0.714	0.214
has_emoji	0.398	0.102
has_latin	0.598	0.098
n_emoji	0.403	0.097
has_newline	0.566	0.066
has_hashtag	0.556	0.056
laugh_emoji	0.553	0.053
has_mention	0.551	0.051
has_url	0.526	0.026
very_short	0.477	0.023
has_question	0.513	0.013
has_quote	0.510	0.010
has_exclaim	0.507	0.007
has_elongation	0.497	0.003
has_laughter	0.501	0.001

For taking out the cue, I took out the columns n_chars, laugh_emoji, n_emoji has_hashtag, n_tokens, has_mention, starts_with_mention, has_newline, ends_with_emoji, has_url, has_repeat_punct, has_latin, has_elongation has_laughter, has_exclaim, has_emoji, very_short, has_quote, has_question

- **Code-mixed (English) words:** present in 15.8% of posts, 70.7% of them sarcastic (lift 2.0). This is the most valuable cue because it is both strong and common.
- **Hashtags:** 96.7% sarcastic (lift 2.5). Partly a label giveaway because of #sarcastic
- **Newlines, mentions, URLs:** 90–93% sarcastic (lift about 2.3–2.5). They are strong but probably collection artifacts, not real sarcasm.
- **Exclamation mark:** 93.1% sarcastic (lift 2.3), but only 72 posts.
- **Question mark and quotes:** about 55% sarcastic (lift 1.35). The signal is modest but significant.
- **Laugh emojis:** 48.1% sarcastic (lift 1.30) and present in 36.5% of posts.

Emojis depend on the type

- **Any emoji (lift 0.53):** 34.8% of posts with an emoji are sarcastic, against 65.3% without. Emoji presence points toward non-sarcastic.
- **Laugh emoji (lift 1.30):** this points toward sarcastic.
- **Conclusion:** removing all emojis would hide this split. Keep them as separate tokens and group them by sentiment
- **No useful signal, so safe to normalize**
- **Repeated punctuation:** lift 0.99, p = 0.87. No information.
- **Laughter patterns (ہاہاہا):** lift 1.17, p = 0.27, only 73 posts. Not significant.
- **Elongated letters:** lift 0.76, p = 0.036, only 116 posts. After a correction for testing 14 cues, this is not significant either.
- **What to do:** collapse the repetition and keep a flag token, since it costs nothing.

Length is a strong cue

- Very short posts are only 2.8% sarcastic (lift 0.067). They are almost always non-sarcastic, such as greetings or religious phrases.
- Keep text length as a feature.

Caveats to state

- Each cue is tested alone. Hashtag, mention, URL and newline likely occur together, so their effects overlap.
- With about 10,000 posts, a small p-value only means a difference exists. Judge size by lift and coverage.
- An association with the label doesn't make a cue a real sarcasm marker.

- Whether each cue improves the model needs the ablation tables (macro-F1 with the cue kept vs removed).

3.Classical baseline

I have trained the all the model of classical baseline classifier (logistic regression, SVM and word

	features	val_macro_F1
0	char (2-5 char_wb)	0.8207
1	word+char (1, 2)	0.8194
2	word (1, 2)	0.8096
3	word (1, 3)	0.8083
4	word (1, 1)	0.7984

saved outputs/results/ablation_stage2_features.csv (5 rows)

	classifier	C	val_macro_F1
0	svm	0.1	0.8220
1	lr	1.0	0.8217
2	lr	3.0	0.8194
3	svm	0.3	0.8175
4	lr	10.0	0.8085
5	svm	1.0	0.8039

saved outputs/results/ablation_stage3_classifier.csv (6 rows)

```
Best classical model on validation: TF-IDF (word+char) LR/SVM
[TF-IDF (word+char) LR/SVM] TEST macro-F1 = 0.8194 | acc = 0.8286
              precision    recall  f1-score   support

non-sarcastic     0.8301    0.8921    0.8600      1298
    sarcastic     0.8259    0.7370    0.7789       901

     accuracy                         0.8286      2199
    macro avg     0.8280    0.8146    0.8194      2199
 weighted avg     0.8284    0.8286    0.8268      2199

saved outputs/reports/tf_idf_word_char_lr_svm_classification_report.json
saved outputs/models/tf_idf_word_char_lr_svm.pkl  (3.0 MB)
[TF-IDF + cues] TEST macro-F1 = 0.8213 | acc = 0.8304
              precision    recall  f1-score   support
```

The best classical model we found is TF-IDF

4.Neural network

```
[BiGRU (cues kept)] TEST macro-F1 = 0.7972 | acc = 0.8026
              precision    recall  f1-score   support

non-sarcastic     0.8429    0.8182    0.8303      1298
    sarcastic     0.7487    0.7802    0.7641       901

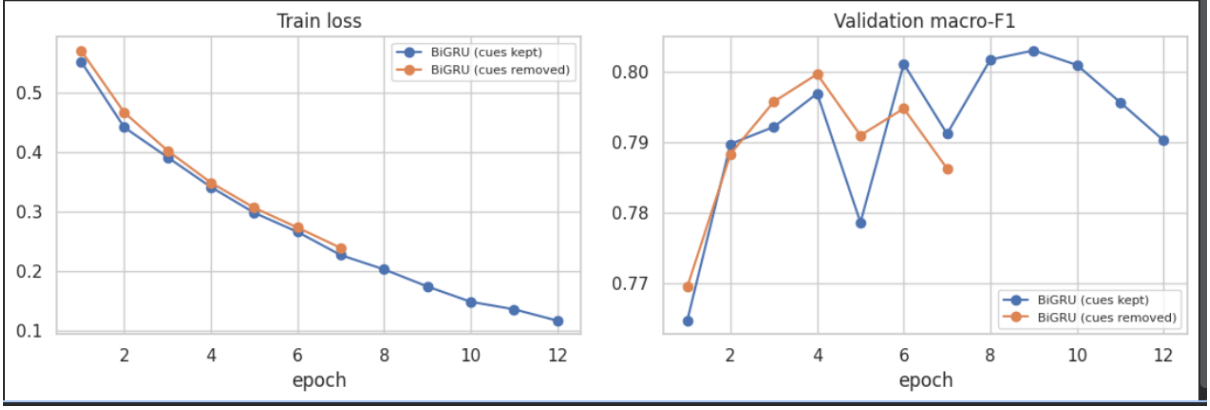
     accuracy                         0.8026      2199
    macro avg     0.7958    0.7992    0.7972      2199
 weighted avg     0.8043    0.8026    0.8032      2199

saved outputs/reports/bigru_cues_kept_classification_report.json
[BiGRU (cues removed)] TEST macro-F1 = 0.8027 | acc = 0.8081
              precision    recall  f1-score   support

non-sarcastic     0.8465    0.8243    0.8353      1298
    sarcastic     0.7561    0.7847    0.7702       901

     accuracy                         0.8081      2199
    macro avg     0.8013    0.8045    0.8027      2199
 weighted avg     0.8095    0.8081    0.8086      2199
```

```
saved outputs/reports/bigru_cues_removed_classification_report.json
saved outputs/figures/08_bigru_training_curves.png
```



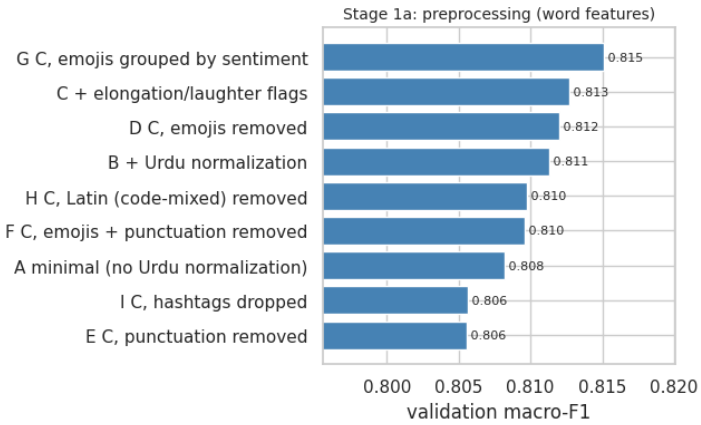
5.Transfarmer

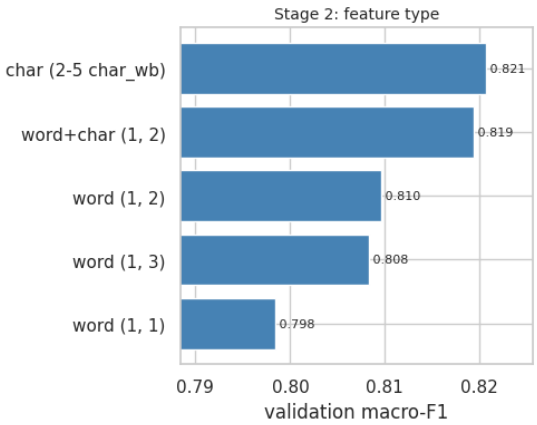
```
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (
===  XLM-R base (emojis kept)
Loading weights: 100% ████████████████████████████████ 197/197 [00:00<00:00, 859.21it/s]
[transformers] XLMRobertaForSequenceClassification LOAD REPORT from: xlm-roberta-base
Key                          | Status     |
```

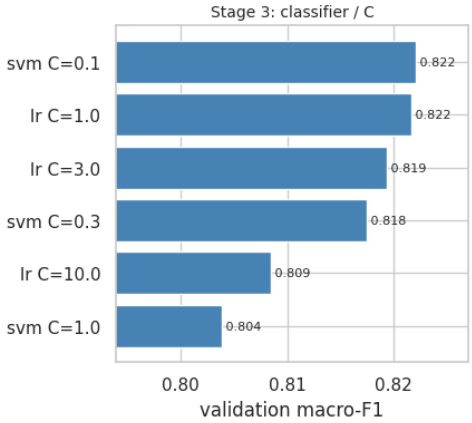
First, I train the model on 4 epoch then train it to 50 epoch

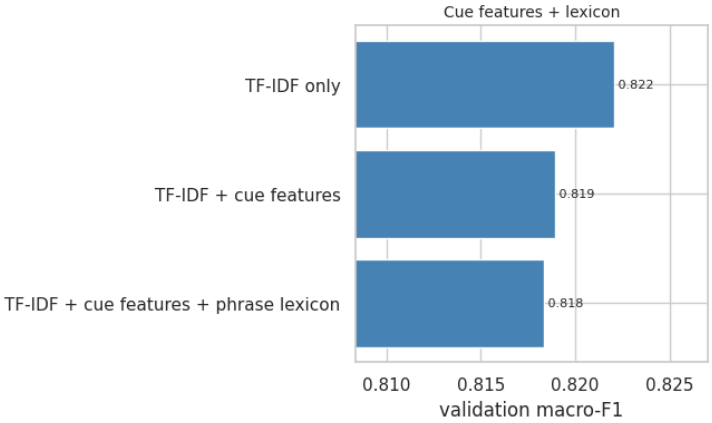
6.Ablation study/ preprocessing/feature

	A	B	C	D
1	preprocessing	features	val_macro_F1	
2	G C, emojis grouped by sentiment	word	0.81509308	
3	C + elongation/laughter flags	word	0.812684015	
4	D C, emojis removed	word	0.812003314	
5	B + Urdu normalization	word	0.811294935	
6	H C, Latin (code-mixed) removed	word	0.809756266	
7	F C, emojis + punctuation removed	word	0.809604657	
8	A minimal (no Urdu normalization)	word	0.808215026	
9	I C, hashtags dropped	word	0.805667321	
10	E C, punctuation removed	word	0.805531555	
11				
12				
13				
14				







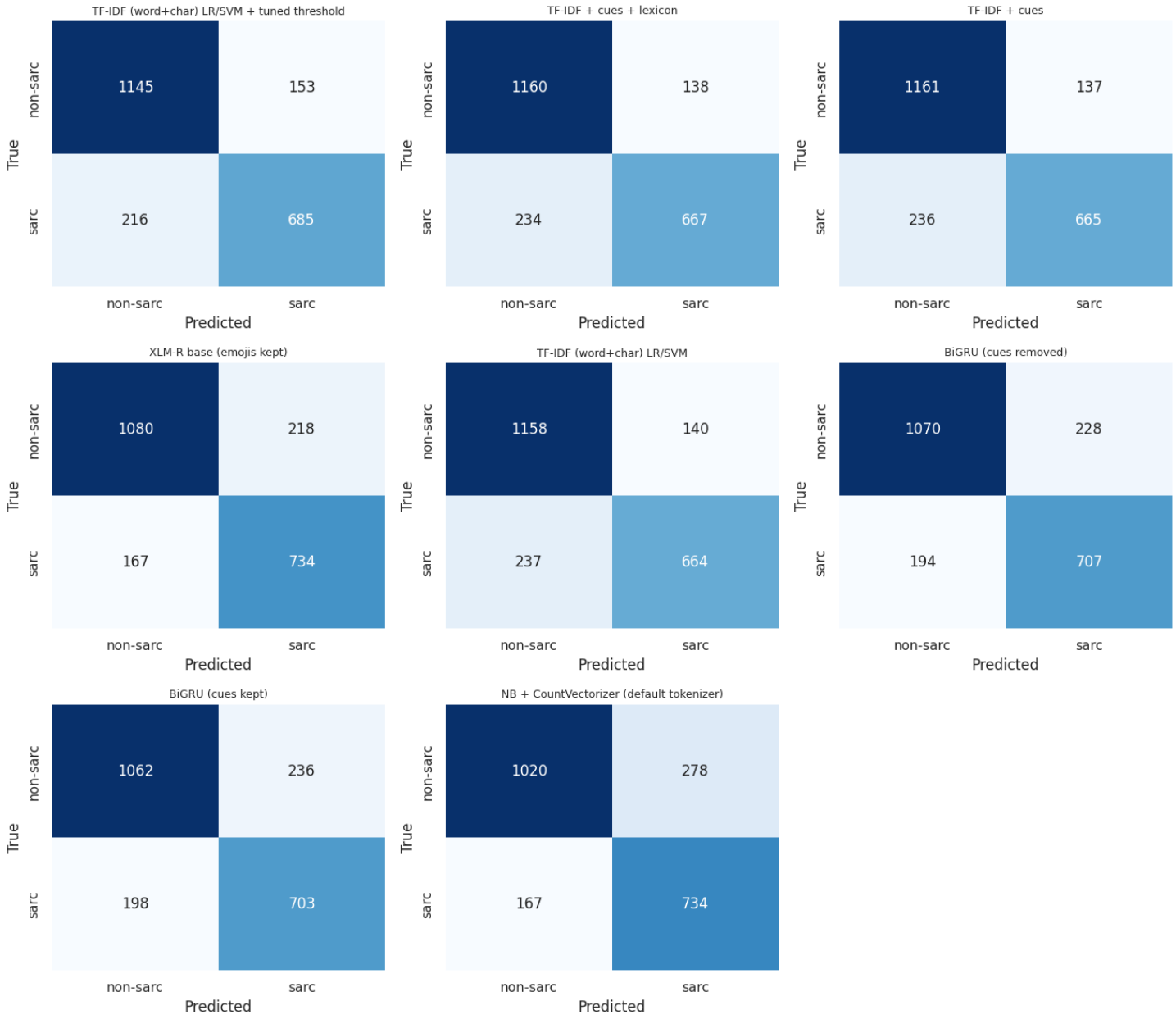


- **Normalized vs minimal text:** normalization improved the score from 0.8082 to 0.8113, and the elongation and laughter flags took it to 0.8127. The gain is small but points the right way.
- **Emojis:**
    - Removing them changed almost nothing (0.8127 to 0.8120).
    - Grouping them by sentiment scored highest of all (0.8151).
    - Emojis help mainly when grouped.
- **Punctuation:** removing it gave the lowest score (0.8055, about −0.7 points), so punctuation carries some signal.
- **Hashtags:** dropping them lowered the score to 0.8057 (about −0.7 points), so hashtags help.
- **Code-mixed (English) words:** removing them lowered the score to 0.8098 (about −0.3 points), a small loss.
- **Word vs character features:**
    - Character n-grams (2–5) scored 0.8207, against 0.8096 for word bigrams.
    - Word plus character scored 0.8194, essentially the same as character alone.
    - Bigrams beat unigrams by about +1.1 points, and trigrams added nothing.
    - Character n-grams probably help with Urdu spelling variation.
- **Cue features and phrase lexicon (bonus):** TF-IDF alone scored 0.8220, and adding the cue features (0.8190) and the lexicon (0.8184) did not help. This is a negative result: the n-grams already capture those cues.
- **Solid conclusions:** character features beat word features, bigrams beat unigrams, and removing punctuation or hashtags hurts. The other differences are too small to claim.

7.Evaluation

7.1 Accuracy, Precision, Recall, F1 for each class, Macro-F1

	val_macro_F1	accuracy	P_0	R_0	F1_0	P_1	R_1	F1_1	macro_F1
TF-IDF (word+char) LR/SVM + tuned threshold	0.8258	0.8322	0.8413	0.8821	0.8612	0.8174	0.7603	0.7878	0.8245
TF-IDF + cues + lexicon	0.8184	0.8308	0.8321	0.8937	0.8618	0.8286	0.7403	0.7819	0.8219
TF-IDF + cues	0.8190	0.8304	0.8311	0.8945	0.8616	0.8292	0.7381	0.7810	0.8213
XLM-R base (emojis kept)	0.8234	0.8249	0.8661	0.8320	0.8487	0.7710	0.8147	0.7922	0.8205
TF-IDF (word+char) LR/SVM	0.8220	0.8286	0.8301	0.8921	0.8600	0.8259	0.7370	0.7789	0.8194
BiGRU (cues removed)	0.7996	0.8081	0.8465	0.8243	0.8353	0.7561	0.7847	0.7702	0.8027
BiGRU (cues kept)	0.8030	0.8026	0.8429	0.8182	0.8303	0.7487	0.7802	0.7641	0.7972
NB + CountVectorizer (default tokenizer)	0.7985	0.7976	0.8593	0.7858	0.8209	0.7253	0.8147	0.7674	0.7942



7.2 Error analysis

```
Best model on validation: TF-IDF (word+char) LR/SVM + tuned threshold
369 errors out of 2199 test posts (16.8%)
{'false negative': 216, 'false positive': 153}
{'emoji': 311, 'negation': 114, 'code-mixed': 41, 'long': 38, 'question': 22,
saved outputs/results/error_analysis.csv  (16 rows)
saved outputs/results/error_analysis_all_errors.csv  (369 rows)
```